

Leaky Forms: A Study of Email and Password Exfiltration Before Form Submission

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Leaky Forms: A Study of Email and Password Exfiltration Before Form Submission

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Web users enter their email addresses into online forms for a variety of reasons, including signing in or signing up for a service or subscribing to a newsletter. While enabling such functionality, email addresses typed into forms can also be collected by third-party scripts even when users change their minds and leave the site without submitting the form. Email addresses—or identifiers derived from them—are known to be used by data brokers and advertisers for cross-site, cross-platform, and persistent identification of potentially unsuspecting individuals. In order to find out whether access to online forms is misused by online trackers, we present a measurement of email and password collection that occurs before the form submission on the top 100,000 websites. We evaluate the effect of user location, browser configuration, and interaction with consent dialogs by comparing results across two vantage points (EU/US), two browser configurations (desktop/mobile), and three consent modes. Our crawler finds and fills email and password fields, monitors the network traffic for leaks, and intercepts script access to filled input fields. Our analyses show that users' email addresses are exfiltrated to tracking, marketing and analytics domains before form submission and without giving consent on 1,844 websites in the EU crawl and 2,950 websites in the US crawl. While the majority of email addresses are sent to known tracking domains, we further identify 41 tracker domains that are not listed by any of the popular

blocklists. Furthermore, we find incidental password collection on 52 websites by third-party session replay scripts.

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